SimpleFT8 — Two Antennas, One Comparison

Resonant TX antenna + Diversity RX — the actual killer advantage — 20m FT8

Period: 2026-04-20 to 2026-05-11 · 15s cycles evaluated: 34954

Summary:

On 20m, ANT1 (Kelemen DP-201510) operates WITHIN its design band — a top-class resonant antenna.
Yet: ANT2 (gutter) wins in 79-86% of dual-receive cases with average +4 dB advantage.
This is NOT antenna mismatch like on 40m — it's pure polarization and pattern diversity.
Asymmetric advantage: resonant TX (stations hear me) + RX diversity (I hear them back).

Note on setup: ANT1 (Kelemen DP-201510 trap dipole) operates within its design band on 20m — a first-class resonant antenna. ANT2 (house gutter ~1

What do the three modes mean?

Normal (grey): One antenna, no special logic — classic single-antenna setup. Reference baseline.
Diversity Standard (blue): Two antennas. Auto-selects the one with more stations.
Diversity DX (orange): Two antennas. Selects the one with more weak DX signals (below −10 dB).
Rescue (green caps): Stations ANT1 couldn't hear — ANT2 decoded them anyway.

How was it measured?

Setup, time period and a bit of background

The Setup

Measured on 20m FT8 using a FlexRadio FLEX-8400M — two antenna ports, same frequency.
Period: 2026-04-20 to 2026-05-11. Database growing — 20m measured in parallel to 40m.
Cycles: Normal 7037 (7 day/s) | Diversity Standard 13913 (9 day/s) | Diversity DX 14004 (8 day/s).
Note: 20m has half the wavelength of 40m → polarization diversity acts significantly stronger.

Why not just use the daily average?

Good question — I did it that way at first too. But if one day had only 20 cycles and another had 500, the short day would carry the same weight. That's unfair. So all individual values are pooled and averaged directly — regardless of which day. This gives a picture much closer to reality.

What are 'rescued stations' (Rescue)?

Imagine a station transmitting so weakly that ANT1 receives it below −24 dB.
That is the FT8 decoding threshold — below that, decoding is practically impossible.
ANT2 receives the same station with slightly more signal and decodes it anyway.
We call this 'Rescue'. The green caps in the charts show exactly these stations.
Whether you count them or treat them separately — that's up to you. Both values are in this report.

Where does the raw data come from?

SimpleFT8 writes a small Markdown file per hour containing all cycle values.
No pre-computed averages — raw data only. The analysis script calculates everything fresh.
Want to check? statistics/ in the GitHub repo, all public.

Main Results — Comparison Table

20m FT8 · Pooled mean across all measurement points (database still growing)

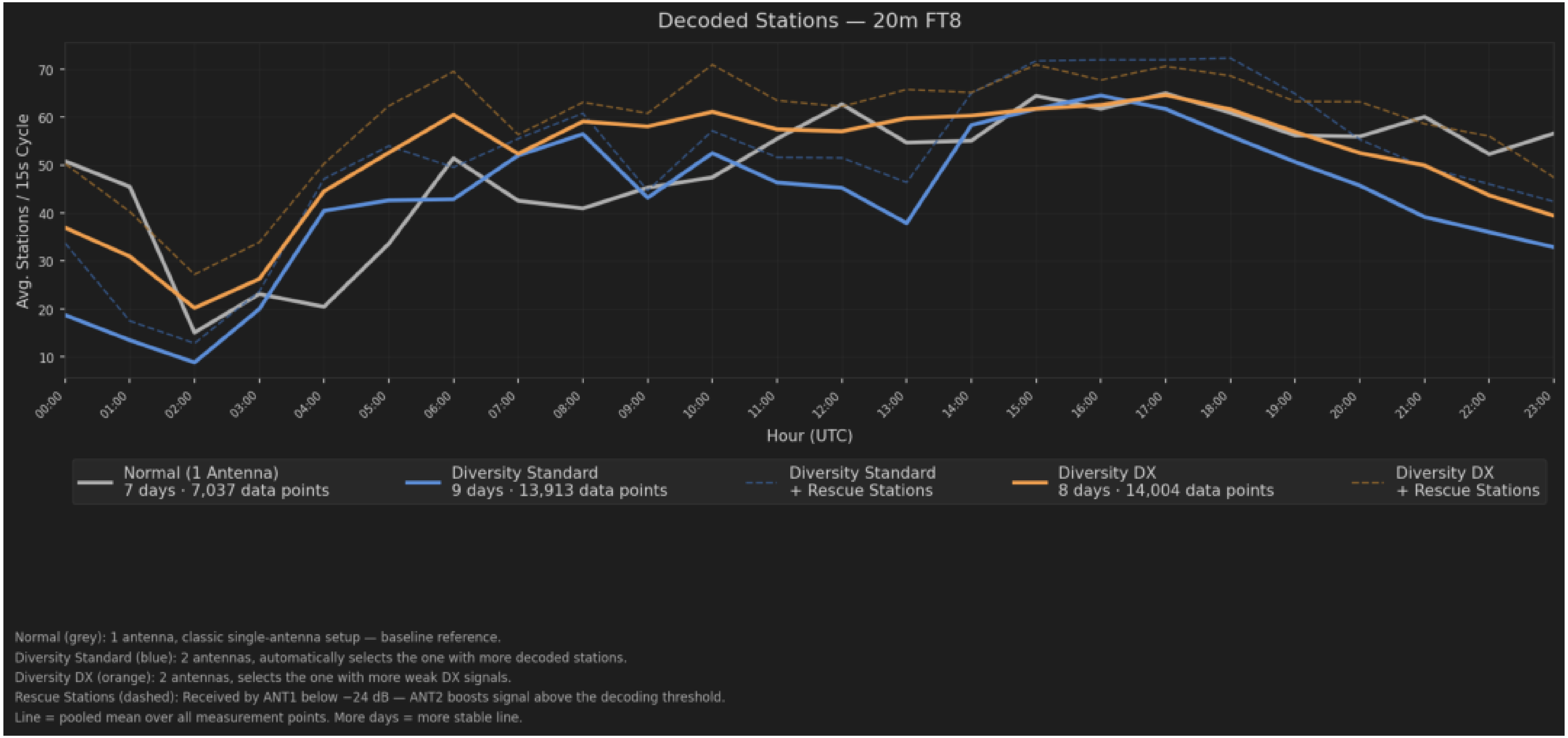
Mode	Avg Sta. /15s Cycle	vs Normal w/o Rescue	vs Normal + Rescue	Rescue only	Meas. Days	Cycles
Normal (1 Antenna	49.9	—	—	—	7	7037
Diversity Standard	46.6	**+-7%**	**+12%**	**+18%**	9	13913
Diversity DX	51.4	**+3%**	**+18%**	**+15%**	8	14004

Avg Sta./15s Cycle: Pooled mean — all measurement points across all days and hours combined. On 20m the database is still thinner than on 40m, so values show more variatio

Important: On 20m ANT1 is RESONANT — the diversity gain does NOT come from antenna mismatch. For stations where BOTH antennas receive, the gutter wins in 79% (Standar

Stations per Hour — all three modes compared

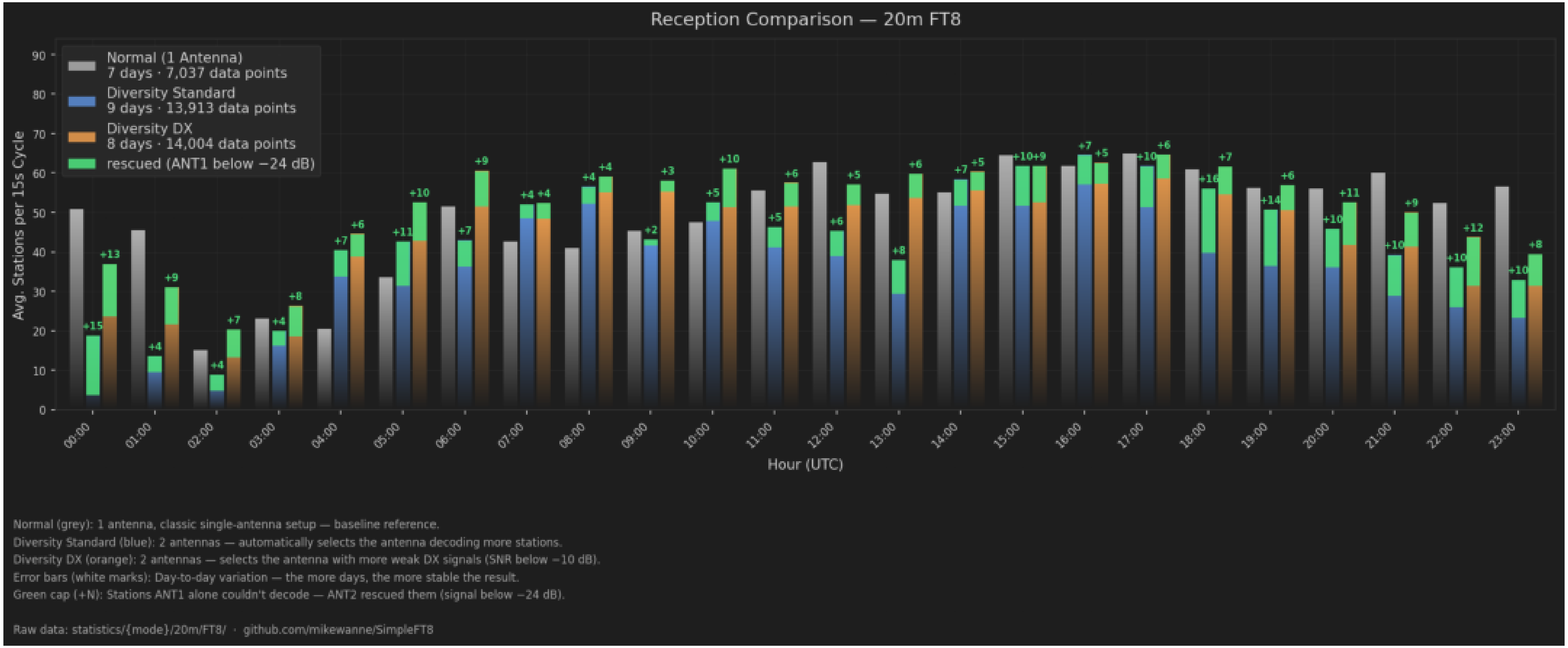
20m FT8 — line = pooled mean over all measurement days. Dashed = + rescue stations.



On 20m the lines are closer together than on 40m — ANT1 is resonant here and already delivers a lot. Yet diversity stays consistently above. The shaded band is wider than on 40m because the database is still thinner — will narrow with more measurement days.

Direct Comparison — bars per hour, three modes side by side

20m FT8 — Normal (grey) | Diversity Standard (blue) | Diversity DX (orange) | Rescue caps (green)



During daytime peak (12-16 UTC) the diversity advantage shows most clearly. At day→night transition (18 UTC) Diversity_DX jumps especially high — DX mode filters for weak signals (SNR<-10dB), and exactly these arrive more often via the quasi-vertical (gutter) at skip-zone edges. White error bars show variation between measurement days.

The Green Caps — count them or not?

Rescue stations: What are they and what do they tell us?

What's this about?

In the diversity chart, small green caps with +N appear above some bars.
These are stations ANT1 couldn't decode — signal below −24 dB.
ANT2 heard and decoded them anyway. Average over the measurement period:
Ø 9.2 stations per hour for Diversity Standard, Ø 7.5/h for Diversity DX.
On 20m the rescue values are typically smaller than on 40m — because ANT1 is resonant here and decodes more signals itself above threshold. The diversity gain comes more from polarization/pattern advantage than from rescue.

Why should you count them in?

Because the QSO is real for the operator — regardless of whether ANT1 or ANT2 made it possible.
These stations would never have made it into the log with a single antenna.
This is not a measurement artifact — it's exactly the point of running a second antenna.
With Rescue: Standard +12%, DX +18% — that's the full picture.

Why might you leave them out?

SNR values fluctuate within the 15 seconds of a cycle — a station just below −24 dB might be above that threshold in the next cycle.
The −24 dB threshold is an empirical value, not a law of nature.
For a clean comparison with other systems, counting only what ANT1 directly decoded is more conservative.

That's why this report always shows both numbers: without Rescue (the conservative value) and with Rescue (what you get in real operation). Ta

What can you take away from this?

Conclusions and what gets measured next

What is clearly visible:

Diversity gain on 20m is SMALLER than on 40m — but that's exactly the point:
On 20m, ANT1 is already a top-class resonant antenna, so 'more' was not to be expected.
Yet the data shows: ANT2 adds on average +-7% to +12% —
and in 79-86% of dual-receive cases, the gutter is even STRONGER than the Kelemen dipole.
Diversity DX delivers 3% — targeted at weak DX signals.
The strongest diversity effect appears at day→night transition (04:00 UTC, +98%) — typically the skip-zone edge where polarization diversity acts

What cannot be said yet:

The 20m database is still significantly thinner than 40m — certain UTC hours (06-09, 20-23) underrepresented.
With very open band (sunspot maximum) the diversity gain might be even higher — so far only moderate conditions measured.
On 15m/10m the effect would theoretically be stronger (Faraday rotation scales with f^2) — but deliberately not recorded (Mike's choice: two band
Whether this works the same on other transceivers — only tested on the FLEX so far.

What comes next:

More measurement days so per-UTC-hour variation decreases.
Targeted measurements 06-09 UTC and 20-23 UTC — closing the data gaps.
Updates run in parallel to the 40m analysis — both reports maintained together.
This report updates automatically whenever new data comes in.

Raw data: statistics/ in the repo · github.com/mikewanne/SimpleFT8 · DA1MHH